

# Soft Mutation in Colloquial Welsh: A Predictive Account

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*Generated by report/report.ts. Do not edit by hand: every tree below is rendered by the program and every stated verdict is asserted at build time — this document cannot disagree with the implementation that backs it.*

**Scope.** Prescriptive standard modern **colloquial** Welsh, with Gareth King’s *Modern Welsh: A Comprehensive Grammar* (3rd ed., 2016) as the normative reference. The account is strictly **predictive**: given a word and its grammatical context, does it exhibit soft mutation (SM)? Real spoken usage is not categorical (Jones, n.d.; Knight et al. 2020) — §7 records where prescription and usage genuinely diverge. The phonological *content* of mutation ( $t \rightarrow d$ ,  $p \rightarrow b$ ,  $c \rightarrow g$ , ...) is taken as opaque throughout; only the *trigger conditions* are at issue. Word-internal mutation under derivational prefixes is out of scope: the predicate ranges over tokens, and prefixed forms are radical lexemes in their own right.

## 1. The heterogeneity thesis

Every current theoretical account, whatever its framework, starts from the same premise: soft mutation has no single cause. Most instances have a lexical or morphosyntactic trigger, while a distinct residue arises under specifically syntactic conditions (Tallerman 2006). Green (2003) states the consensus

most bluntly: trigger environments split into proclitic-triggered and syntax-triggered classes, and *no single generalization covers both* — the full set is arbitrary, partly item-specific, and dialectally variable. Historically this arbitrariness is the residue of reanalysis: mutation was once predictable from pure phonology (the final segment of the preceding word), but by Middle Welsh it could be predicted only from a list of triggering environments, the overwhelming majority being individual lexical items acting on the immediately following word (Willis 2007).

The program therefore implements three subsystems plus a veto layer, each returning its own provenance:

1. **Contact triggers** (lex:\*) — a closed class of words governing a mutation grade on the immediately following dependent.
2. **Gender marking** (gend:\*) — feminine singular nouns after the article and *un*; modifiers agreeing with a feminine singular controller.
3. **Syntactic/positional mutation** (synt:\*) — the XP Trigger Hypothesis plus a small set of designated positions.

One constraint spans all three — the **Trigger Constraint** (Breit 2019): every mutation is licensed by the immediately preceding string element or by a feature borne by the target itself. Nothing acts at a distance. This is the strongest cross-cutting invariant in the literature and it is an invariant of the program: the environment a word is judged in is a fixed-size record about one preceding position plus the word’s own resolved features, never a whole tree.

## 2. Reading the diagrams

Each example is rendered from the program’s tree representation. Child indices appear on the connectors; every leaf is annotated with the program’s verdict and its provenance. (N f sg) gives category and features; lemma= marks a trigger-lexicon key that differs from the citation form (homographs like the two *yn*, the two *ei*); gap:NP is an extraction site. In the Welsh text, ° marks a soft-mutated word. Verdict annotations appear on *every* leaf, so incidental facts (the colloquial mutation of the clause-initial verb, vetoes on function words) are visible alongside the example’s point.

## 3. Contact triggers

The majority case (Willis 2007). Each trigger is a lexical entry governing a grade: full SM; *limited* SM sparing *ll-/rh-* (the article, *un*, predicative *yn*, *mor* — King §9); AM; NM; the *mixed* grade (AM on *c/p/t*, SM elsewhere — King §10); or nothing (absence is data: *pob*, *mewn*, *rhwng*, *efo*, *tair* are stored as non-triggers, not omitted). AM and NM frames matter to an SM predicate because they *pre-empt*: a word governed by an AM trigger does not soft-mutate. Blocking (King §5d) requires no rule — a trigger acts only on the immediately following word, so an intervener simply *is* the preceding element.

**i °dŷ**

‘to a house’ — the preposition i° governs SM on its dependent (King §460)

PP

└<sub>0</sub> i {0other} → radical (no-license)

└<sub>1</sub> NP

└<sub>0</sub> tŷ (N m sg) → SM (lex:i)

**dy °gath**

‘your cat’ — 2sg possessive dy° (King §111)

NP

├<sub>0</sub> dy ⟨Other⟩ immutable → radical (no-license)  
└<sub>1</sub> cath ⟨N f sg⟩ → SM (lex:dy)

### **ei °gath e — ei chath hi**

‘his cat’ (SM) vs ‘her cat’ (AM, hence radical for the SM predicate) — one orthographic word, two lemmas, two grades (King §112)

NP

├<sub>0</sub> ei ⟨Other⟩ lemma=ei.3sgm → radical (no-license)  
├<sub>1</sub> cath ⟨N f sg⟩ → SM (lex:ei.3sgm)  
└<sub>2</sub> e ⟨Other⟩ → radical (no-license)

With lemma ei.3sgf the frame grade is AM: the same tree position yields radical (no-license) — ‘her’ never soft-mutates.

### **fy nghath**

‘my cat’ — fy governs NM, so the SM predicate correctly refuses (King §110)

NP

├<sub>0</sub> fy ⟨Other⟩ → radical (no-license)  
└<sub>1</sub> cath ⟨N f sg⟩ → radical (no-license)

### **mor °fach — mor llawn**

‘so small’ vs ‘so full’ — mor° is limited SM: it spares ll-/rh- (King §105e)

AP

├<sub>0</sub> mor ⟨Adv⟩ immutable → radical (no-license)  
└<sub>1</sub> bach ⟨Adj⟩ → SM (lex:mor)

Substituting llawn (ll-) at the same position yields radical (no-license): the SM-ltd grade has no reflex for ll-/rh-.

### **mor llawn**

‘so full’ — the ll- target resists the limited-SM trigger

AP

├<sub>0</sub> mor ⟨Adv⟩ immutable → radical (no-license)  
└<sub>1</sub> llawn ⟨Adj⟩ → radical (no-license)

### **dwy °gath — tair cath**

‘two cats’ vs ‘three cats’ — dwy° governs SM, tair governs nothing (King §162)

NP

├<sub>0</sub> dwy ⟨Num⟩ → radical (no-license)  
└<sub>1</sub> cath ⟨N f sg⟩ → SM (lex:dwy)

### **tair cath**

‘three cats’ — feminine tair is always followed by the radical (King §162c)

NP

├<sub>0</sub> tair ⟨Num⟩ → radical (no-license)  
└<sub>1</sub> cath ⟨N f sg⟩ → radical (no-license)

### **chwe cheffyl**

‘six horses’ — chwe governs AM, never SM (King §162e)

NP

- |<sub>0</sub> chwe ⟨Num⟩ → radical (no-license)
- |<sub>1</sub> ceffyl ⟨N m sg⟩ → radical (no-license)

### **chwe mlynedd — dwy °flynedd**

‘six years’ (NM via the year-word frame) vs ‘two years’ (plain SM) — one lemma, two frames (King §176)

NP

- |<sub>0</sub> chwe ⟨Num⟩ → radical (no-license)
- |<sub>1</sub> blynedd ⟨N f sg⟩ → radical (no-license)

chwe carries two frames: AM generally, NM restricted to blwydd/blynedd/diwrnod. Neither yields SM. dwy °flynedd (example above) shows the same noun soft-mutating after a genuine SM trigger.

### **hen °ddyn**

‘an old man’ — prenominal adjectives govern SM (King §96)

NP

- |<sub>0</sub> hen ⟨Adj⟩ → radical (no-license)
- |<sub>1</sub> dyn ⟨N m sg⟩ → SM (lex:hen)

### **pob dyn**

‘every man’ — pob is the one prenominal adjective that governs nothing (King §97)

NP

- |<sub>0</sub> pob ⟨Other⟩ → radical (no-license)
- |<sub>1</sub> dyn ⟨N m sg⟩ → radical (no-license)

### **cath neu °gi**

‘a cat or a dog’ — neu° governs SM on the second conjunct (King §512)

NP

- |<sub>0</sub> NP
  - |<sub>0</sub> cath ⟨N f sg⟩ → radical (no-license)
- |<sub>1</sub> neu ⟨Other⟩ → radical (no-license)
- |<sub>2</sub> NP
  - |<sub>0</sub> ci ⟨N m sg⟩ → SM (lex:neu)

### **y °gath neu’r ci**

‘the cat or the dog’ — the article ’r intervenes and blocking follows from string adjacency (King §5d)

NP

- |<sub>0</sub> NP
  - |<sub>0</sub> y ⟨Other⟩ → radical (no-license)
  - |<sub>1</sub> cath ⟨N f sg⟩ → SM (gend:art-fem-sg)
- |<sub>1</sub> neu ⟨Other⟩ → radical (no-license)
- |<sub>2</sub> NP
  - |<sub>0</sub> y ⟨Other⟩ → radical (veto:no-reflex blocks lex:neu)
  - |<sub>1</sub> ci ⟨N m sg⟩ → radical (no-license)

No blocking rule exists in the program: 'r simply IS the preceding element, and its own frame (feminine singular nouns) does not apply to masculine ci.

## 4. The gender subsystem

Welsh marks gender by mutation rather than affixation. Two environments: the definite article (and *un*) mutates feminine singular nouns, sparing *ll-/rh-* (King §28); and modifiers of a feminine singular controller mutate — with *no ll-/rh-* sparing (King §100: *profedigaeth °lem*), so the carve-out is category-sensitive, not phonological (Green 2003). Agreement is implemented as a feature borne by the target (Breit 2019), not as adjacency to the noun: that is what carries adjective chains, and what lets a *prenominal* adjective agree with a head that follows it. Possessor phrases are systematically immune despite string adjacency (Mittendorf & Sadler 2006; Dowle 2024) — in the program this is the *absence of a licensing relation*, derived from the genitive configuration, not a veto.

### y °gath — y llong — y plant

‘the cat / the ship / the children’ — the article mutates feminine singulars only, sparing *ll-/rh-* (King §28)

NP

└<sub>0</sub> y {Other} → radical (no-license)  
└<sub>1</sub> cath {N f sg} → SM (gend:art-fem-sg)

### y llong

‘the ship’ — feminine singular in *ll-* resists the article (King §28 note b)

NP

└<sub>0</sub> y {Other} → radical (no-license)  
└<sub>1</sub> llong {N f sg} → radical (no-license)

### y cathod

‘the cats’ — feminine plurals pattern with masculines: no mutation (King §28)

NP

└<sub>0</sub> y {Other} → radical (no-license)  
└<sub>1</sub> cathod {N f pl} → radical (no-license)

### y °ddau °gi

‘the two dogs, both dogs’ — the article’s second frame mutates the numerals *dau/dwy*, which then trigger in turn (King §29)

NP

└<sub>0</sub> y {Other} → radical (no-license)  
└<sub>1</sub> dau {Num} → SM (lex:y)  
└<sub>2</sub> ci {N m sg} → SM (lex:dau)

### cath °fach

‘a little cat’ — adjectives agree with a feminine singular controller (King §102)

NP

└<sub>0</sub> cath {N f sg} → radical (no-license)  
└<sub>1</sub> AP  
└<sub>0</sub> bach {Adj} → SM (gend:agr-mod)

### **y °ferch °fach °wen**

‘the little white girl’ — the second adjective is not adjacent to the noun, yet mutates: agreement is a feature borne by the target (Breit 2019), not a contact effect

NP

```
└─0 y ⟨Other⟩ → radical (no-license)
└─1 merch ⟨N f sg⟩ → SM (gend:art-fem-sg)
└─2 AP
  └─0 bach ⟨Adj⟩ → SM (gend:agr-mod)
└─3 AP
  └─0 gwyn ⟨Adj⟩ → SM (gend:agr-mod)
```

### **ci mawr coch**

‘a big red dog’ — masculine chains show no mutation anywhere; this datum forces the NP-internal exclusion on the syntactic subsystem (see §5)

NP

```
└─0 ci ⟨N m sg⟩ → radical (no-license)
└─1 AP
  └─0 mawr ⟨Adj⟩ → radical (no-license)
└─2 AP
  └─0 coch ⟨Adj⟩ → radical (no-license)
```

### **y °brif °ddinas**

‘the capital city’ — the PREnominal adjective agrees with a FOLLOWING feminine head; agreement looks rightward within the NP

NP

```
└─0 y ⟨Other⟩ → radical (no-license)
└─1 prif ⟨Adj⟩ → SM (gend:agr-mod)
└─2 dinas ⟨N f sg⟩ → SM (lex:prif)
```

### **cath merch**

‘a girl’s cat’ — the possessor is immune despite string adjacency to a feminine noun (Mittendorf & Sadler 2006; Dowle 2024)

NP

```
└─0 cath ⟨N f sg⟩ → radical (no-license)
└─1 NP
  └─0 merch ⟨N f sg⟩ → radical (no-license)
```

Derived, not stipulated: the genitive configuration [NP N NP] yields relation=possessor, and possessors are simply outside every licensing relation.

### **canol y °dre**

‘the town centre’ — immunity applies to the possessor boundary, not to the possessor’s inside: the article still fires within it

NP

```
└─0 canol ⟨N m sg⟩ → radical (no-license)
└─1 NP
  └─0 y ⟨Other⟩ → radical (no-license)
  └─1 tre ⟨N f sg⟩ → SM (gend:art-fem-sg)
```

## 5. Syntactic and positional mutation

The theoretically contested residue. Two families competed: Case-based accounts, on which direct object mutation (DOM) realizes accusative Case (Zwicky 1984; Roberts 1997, 2005), and the configurational **XP Trigger Hypothesis** (Harlow 1989; Borsley 1997; Tallerman 2006), on which any phrasal category triggers SM on the word immediately following its right edge. The Case account is generally judged to both overgenerate and undergenerate (Borsley 1997; Tallerman 2006): mutation after subjects hits non-accusative material (*dim*, adverbs), and objects of nonfinite verbs mutate exactly when a phrase intervenes (Green 2003) — facts the XPTH derives for free. The program implements the XPTH geometrically: a phrasal node (never a clause) whose right edge immediately precedes the target and c-commands it licenses `synt: xp-edge`. Extraction gaps count as phrases (Willis 2007 documents the transparency lineage); following Dowle (2024) we posit no other empty categories.

Two implementation findings are worth recording as results. First, raw XPTH geometry overgenerates **NP-internally**: in a masculine chain like *ci mawr coch*, AP(*mawr*) ends immediately before *coch* and c-commands it, yet *coch* is radical. The program therefore excludes candidates whose join node with the target is an NP — Tallerman’s ‘complement’ condition operationalized, and exactly Breit’s (2019) division of labor: NP-internal mutation belongs to the gender subsystem alone. Second, clause-initial finite-verb mutation (`synt: v1-*`) is **not XPTH-derivable even in principle**: nothing precedes a clause-initial verb, and the negative variant surfaces as *mixed* mutation (AM on *c/p/t*) where the XPTH only ever produces SM. It is particle-drop residue — affirmative *fe°/mi°*, interrogative *a°*, negative *ni* (mixed) were deleted, and their mutations remained (King §481 note; Green 2003 on *ni*). The grade tracks the vanished particle exactly. In formal literary Welsh the affirmative default is particle-less and radical, so this entire rule is register-specific (King §11d) — encoded as designated positions, it is a data toggle, not a theory change. Imperatives resist both v1 mutation and *neu°* (King §512), which the category system encodes directly.

### Gwelodd Mair °dŷ

‘Mair saw a house’ — direct object mutation: the subject NP’s right edge licenses, not the verb (Harlow 1989; Borsley 1997; Tallerman 2006)

```
S
├0 gweld ⟨V⟩ → SM (synt:v1-aff)
├1 NP
│  └0 Mair ⟨N⟩ immutable → radical (no-license)
└2 NP
   └0 tŷ ⟨N m sg⟩ → SM (synt: xp-edge)
```

The verb also carries colloquial v1 SM (°Welodd) — see the v1 examples below; the verdict annotation on the verb reflects this.

### Roedd dyn wedi prynu beic

‘A man had bought a bike’ — the object of a NON-finite verb stays radical: no phrase edge precedes it

```
S
├0 bod ⟨V⟩ → radical (veto:no-reflex blocks synt:v1-aff)
├1 NP
│  └0 dyn ⟨N m sg⟩ → radical (no-license)
├2 wedi ⟨Prt⟩ → radical (veto:no-reflex blocks synt: xp-edge)
└3 VNP
   └0 prynu ⟨Vnoun⟩ → radical (no-license)
```

└<sub>1</sub> NP  
└<sub>0</sub> beic (N m sg) → radical (no-license)

### Roedd dyn wedi prynu yn y °dre °feic

‘A man had bought, in town, a bike’ — the decisive datum for the configurational account: an intervening PP restores mutation on the nonfinite object (Green 2003)

S  
└<sub>0</sub> bod (V) → radical (veto:no-reflex blocks synt:v1-aff)  
└<sub>1</sub> NP  
| └<sub>0</sub> dyn (N m sg) → radical (no-license)  
└<sub>2</sub> wedi (Prt) → radical (veto:no-reflex blocks synt:xp-edge)  
└<sub>3</sub> VNP  
| └<sub>0</sub> prynu (Vnoun) → radical (no-license)  
| └<sub>1</sub> PP  
| | └<sub>0</sub> yn (Other) lemma=yn.loc → radical (no-license)  
| | └<sub>1</sub> NP  
| | | └<sub>0</sub> y (Other) → radical (no-license)  
| | | └<sub>1</sub> tre (N f sg) → SM (gend:art-fem-sg)  
| └<sub>2</sub> NP  
| └<sub>0</sub> beic (N m sg) → SM (synt:xp-edge)

### Beic °brynnodd y °ddynes

‘It was a bike the woman bought’ — the fronted object has nothing before it and stays radical; a Case-based account must stipulate this (Tallerman 2006 contra Roberts 1997)

S  
└<sub>0</sub> NP  
| └<sub>0</sub> beic (N m sg) → radical (no-license)  
└<sub>1</sub> prynu (V) → SM (synt:xp-edge)  
└<sub>2</sub> NP  
| └<sub>0</sub> y (Other) → radical (no-license)  
| └<sub>1</sub> dynes (N f sg) → SM (gend:art-fem-sg)

### Pwy °welodd °ddraig?

‘Who saw a dragon?’ — the extraction gap occupies the subject position and counts as a phrase edge; the verb’s own SM (°welodd < gwelodd) comes from the fronted wh-phrase’s edge

S  
└<sub>0</sub> NP  
| └<sub>0</sub> pwy (Other) → radical (no-license)  
└<sub>1</sub> gweld (V) → SM (synt:xp-edge)  
└<sub>2</sub> gap:NP  
└<sub>3</sub> NP  
| └<sub>0</sub> draig (N f sg) → SM (synt:xp-edge)

### Rhaid i Emrys °fynd

‘Emrys must go’ — the PP’s right edge c-commands and licenses the verbal noun; King files this under ‘sentence construction’ (§5e: unblockable), for the XPTH it is the same rule as DOM

S  
└<sub>0</sub> rhaid (N) → radical (no-license)



```

├1 PP
|   ├──0 i {Other} → radical (no-license)
|   └──1 NP
|       └──0 Emrys {N} immutable → radical (veto:no-reflex blocks lex:i)
└2 VNP
    └──0 mynd {Vnoun} → SM (synt:xp-edge)

```

### °Golles i'r tocyn

‘I lost the ticket’ — DOM lands on the first WORD of the object; here that is the article (no SM reflex), so the noun stays radical

```

S
├0 colli {V} → SM (synt:v1-aff)
├1 NP
|   ├──0 i.pron {Other} → radical (no-license)
└2 NP
    ├──0 y {Other} → radical (veto:no-reflex blocks synt:xp-edge)
    └──1 tocyn {N m sg} → radical (no-license)

```

### °Ddylset ti °ddim

‘You shouldn’t’ — two mutations, two subsystems: mixed particle-residue on the negative verb, subject-edge SM on dim (King §§10, 11a)

```

S (neg)
├0 dylu {V} → SM (synt:v1-neg-mixed)
├1 NP
|   ├──0 ti {Other} immutable → radical (no-license)
└2 dim {Prt} → SM (synt:xp-edge)

```

### Pharith hi °ddim

‘It won’t last’ — the negative v1 grade is MIXED: AM on p-, so no soft mutation; the grade tracks the dropped particle ni (King §10)

```

S (neg)
├0 para {V} → radical (no-license)
├1 NP
|   ├──0 hi {Other} → radical (no-license)
└2 dim {Prt} → SM (synt:xp-edge)

```

### os daw e

‘if he comes’ — the subordinator occupies clause-initial position, so the verb is not v1 and stays radical (King §502)

```

S
├0 os {Other} → radical (no-license)
├1 dod {V} → radical (no-license)
└2 NP
    └──0 e {Other} → radical (no-license)

```

### °Ddwyr °flynedd yn ôl aeth hi adre

‘Two years ago she went home’ — adverbial NPs mutate at their first word (King §11b); note the geometrically identical fronted object (dom-fronted) does NOT — the difference is authored, not derived

```

S
├0 NP (adverbial)
│   ├──0 dwy ⟨Num⟩ → SM (synt:adv-np)
│   ├──1 blynedd ⟨N f sg⟩ → SM (lex:dwy)
│   └2 yn ôl ⟨Adv⟩ → radical (no-license)
├1 mynd ⟨V⟩ → radical (veto:no-reflex blocks synt:xp-edge)
└2 NP
    └0 hi ⟨Other⟩ → radical (no-license)

```

### Dewch, °blant!

‘Come, children!’ — vocative mutation (King §11c); the imperative verb itself stays radical: imperatives resist v1 mutation

```

S
├0 dod ⟨Vimp⟩ → radical (no-license)
└1 NP (vocative)
    └0 plant ⟨N m pl⟩ → SM (synt:vocative)

```

## 6. Vetoes

Two ways a word can be exempt no matter what licenses it: a per-lexeme immutability flag (King §12: fixed mutations like *beth*; personal names; g-initial loanwords; miscellaneous grammatical words), and the absence of any SM reflex for the word’s initial (vowels, *n-*, *s-*, *ch-*, *ff-* ..., King §5a). The two are distinguished in provenance because they are different theoretical claims: one is lexical listing, the other phonological vacuity.

### dy gêm

‘your game’ — g-initial loanwords are immutable even under a live trigger (King §12e)

```

NP
├0 dy ⟨Other⟩ immutable → radical (no-license)
└1 gêm ⟨N f sg⟩ immutable → radical (veto:immutable blocks lex:dy)

```

### i Dafydd

‘to Dafydd’ — personal names do not mutate in the modern language (King §12d, §36)

```

PP
├0 i ⟨Other⟩ → radical (no-license)
└1 NP
    └0 Dafydd ⟨N⟩ immutable → radical (veto:immutable blocks lex:i)

```

### i ysgol

‘to school’ — vowel-initial words have no SM reflex; the question does not arise (King §5a)

```

PP
├0 i ⟨Other⟩ → radical (no-license)
└1 NP
    └0 ysgol ⟨N f sg⟩ → radical (veto:no-reflex blocks lex:i)

```

## 7. Contested territory

A predictive account must say where prescription itself is unstable. These are flagged in the program’s contested registry; test failures in these contexts are reported as *contested*, not wrong. Corpus context: overall prescriptive application in spontaneous speech runs near 74%, no speaker is categorical, gender agreement on adjectives applies barely above chance (54%), and the particle-less finite-verb SM is the most categorical trigger measured (95%) — consistent with relexicalization of the mutated verb forms (Jones, n.d.; Knight et al. 2020).

- **euphony:s-d** — d- often fails to mutate after preceding -s: nos da, wythnos diwetha (King §12e note). King treats as euphonic override of gend:agr-mod.
- **am-erosion** — AM after â/gyda/tri frequently realized as SM or radical colloquially (King p.12; corpus: gyda 2.94% AM). We encode King’s prescription (AM ⇒ no SM), but SM here is attested vernacular.
- **immutable-variation** — lle, byth: immutability ‘usage varies’ (King §12b).
- **v1-neg-sm** — Negative inflected verbs prescriptively take mixed mutation, but plain SM is common speech (King §10: ‘more a feature of the literary language’). We encode mixed.
- **placenames** — Non-Welsh placenames prescriptively immutable (i Birmingham), but i Firmingham ‘not uncommon’ in speech (King §12c). After yn ‘in’, foreign names ‘even more chaotic’: yng Nghamden / yn Gamden / yn Camden all heard (King §472).
- **nm-blynedd** — NM of blynedd/blwydd after chwe and wyth ‘sometimes followed by the radical’: chwe blynedd, wyth blynedd (King §176 note a).
- **yn-loc-sm** — NM after yn ‘in’ is ‘precarious at best’ in speech; if any mutation heard it is usually SM (yn °Fangor, yn °Geredigion), though SM of G- names is resisted (yn Gogledd Cymru) and bare radical is common (yn Bangor). Formal written language allows none of these (King §472). Fixed exception either way: yn °Gymraeg takes SM even in the literary language (blocked NM from a lost article).
- **tan-conj** — tan as a conjunction (tan iddo ffonio) ‘certainly used’ in some areas but ‘some speakers regard this as substandard’; standard is nes (King §467).
- **cyn-finite** — cyn + inflected verb (cyn daeth e adre) ‘regarded as substandard’ for cyn iddo °ddod adre (King §509).
- **tra-grade** — Degree adverb tra: King marks tra° (SM, §95); the literary tradition prescribes AM (tra chyflym). We encode King’s SM.
- **am-dialect-mn** — In some dialects AM extends to radical m-/n- (m > mh, n > nh), ‘regarded as non-standard’ (King App A p.488). Irrelevant to the SM predicate but a pipeline hazard: mh-/nh-surface forms are not always NM.

## 8. The full trigger lexicon

Every contact frame in the program, with its King reference. A lemma may carry several frames (y, *chwe*); none frames record deliberate non-triggers.

| Lemma | Grade | Target conditions | Source |
|-------|-------|-------------------|--------|
| am    | SM    | —                 | §448   |
| ar    | SM    | —                 | §449   |
| at    | SM    | —                 | §450   |

| Lemma    | Grade | Target conditions                           | Source                                                                             |
|----------|-------|---------------------------------------------|------------------------------------------------------------------------------------|
| cyn.prep | none  | —                                           | §451 (no superscript; contrast cyn- prefix and cyn.equ)                            |
| dan      | SM    | —                                           | §452                                                                               |
| o dan    | SM    | —                                           | §452                                                                               |
| oddidan  | SM    | —                                           | §452                                                                               |
| dros     | SM    | —                                           | §453                                                                               |
| tros     | SM    | —                                           | §453 (literary variant)                                                            |
| drost    | SM    | —                                           | §453 (spoken variant)                                                              |
| efo      | none  | —                                           | §454 (N ‘with’)                                                                    |
| gan      | SM    | —                                           | §455                                                                               |
| gyn      | SM    | —                                           | §455 (spoken variant of gan)                                                       |
| ger      | none  | —                                           | §456                                                                               |
| gerbron  | none  | —                                           | §456                                                                               |
| gyda     | AM    | —                                           | §457; erosion<br>CONTESTED<br>(am-erosion)                                         |
| heb      | SM    | —                                           | §458                                                                               |
| hyd      | SM    | —                                           | §459                                                                               |
| i        | SM    | —                                           | §460                                                                               |
| o        | SM    | —                                           | §9                                                                                 |
| tan      | SM    | —                                           | §467                                                                               |
| trwy     | SM    | —                                           | §9                                                                                 |
| drwy     | SM    | —                                           | §9                                                                                 |
| wrth     | SM    | —                                           | §9                                                                                 |
| â        | AM    | —                                           | §447 (‘optionally causes AM’) CONTESTED<br>(am-erosion)                            |
| tua      | AM    | —                                           | App A                                                                              |
| yn.loc   | NM    | —                                           | App A, §472                                                                        |
| a.rel    | SM    | —                                           | §481                                                                               |
| dacw     | SM    | —                                           | App A                                                                              |
| dyma     | SM    | —                                           | App A                                                                              |
| dyna     | SM    | —                                           | App A                                                                              |
| fe.prt   | SM    | —                                           | App A §213                                                                         |
| mi.prt   | SM    | —                                           | App A §213                                                                         |
| yma      | SM    | —                                           | App A                                                                              |
| yna      | SM    | —                                           | App A                                                                              |
| neu      | SM    | cat ∈ {N, Adj, Vnoun, Num, Adv, Prt, Other} | §512; SM cancelled before an imperative (Arhoswch neu dewch)<br>— hence V excluded |
| pan      | SM    | —                                           | App A                                                                              |

| Lemma   | Grade  | Target conditions                 | Source                                                                |
|---------|--------|-----------------------------------|-----------------------------------------------------------------------|
| yn.pred | SM-ltd | cat ∈ {N, Adj, Num}               | App A ('not ll-, rh-');<br>numerals in age/time:<br>yn °dair oed §176 |
| go      | SM     | —                                 | §95                                                                   |
| pur     | SM     | —                                 | §95                                                                   |
| rhy     | SM     | —                                 | §95                                                                   |
| tra     | SM     | —                                 | §95 (King marks tra°;<br>literary AM tradition)<br>CONTESTED          |
| reit    | SM     | —                                 | §95                                                                   |
| mor     | SM-ltd | —                                 | §105(e), App A                                                        |
| cyn.equ | SM     | —                                 | §105 (cyn °ddued)                                                     |
| na.than | AM     | —                                 | §103 (na <sup>h</sup> , nag before<br>vowels)                         |
| fatha   | AM     | —                                 | §105 (colloquial 'like')                                              |
| fawr    | SM     | —                                 | §105(d) (fawr °gallach)                                               |
| lawn    | SM     | —                                 | §105(a) (°lawn mor<br>°grac)                                          |
| y       | SM-ltd | cat ∈ {N}; gender=f;<br>number=sg | §28; note (b) ll-/rh-<br>resist                                       |
| y       | SM     | cat ∈ {Num}; lexemes:<br>dau/dwy  | §29 (y °ddau, y °ddwy)                                                |
| un      | SM-ltd | cat ∈ {N}; gender=f;<br>number=sg | §162(a)                                                               |
| fy      | NM     | —                                 | §110                                                                  |
| dy      | SM     | —                                 | §111                                                                  |
| ei.3sgm | SM     | —                                 | §112                                                                  |
| ei.3sgf | AM     | —                                 | §112; colloquial<br>omission CONTESTED<br>(am-erosion)                |
| ein     | none   | —                                 | §113 (h- before vowels<br>only)                                       |
| eich    | none   | —                                 | §113                                                                  |
| eu      | none   | —                                 | §113 (h- before vowels<br>only)                                       |
| 'w.3sgm | SM     | —                                 | §112 (i'w °frawd)                                                     |
| 'w.3sgf | AM     | —                                 | §112                                                                  |
| 'w.3pl  | none   | —                                 | §112                                                                  |
| hen     | SM     | —                                 | §96(a)                                                                |
| prif    | SM     | —                                 | §96(a)                                                                |
| ambell  | SM     | —                                 | §96(a), §116                                                          |
| holl    | SM     | —                                 | §96(a)                                                                |
| pob     | none   | —                                 | §97 (the ONLY<br>prenominal adjective<br>that does not trigger<br>SM) |

| Lemma   | Grade | Target conditions                       | Source                                                               |
|---------|-------|-----------------------------------------|----------------------------------------------------------------------|
| pa      | SM    | —                                       | §96(b)                                                               |
| pwyl.S  | SM    | —                                       | §96(b) (S variant of pa)                                             |
| cyn-    | SM    | —                                       | §96(c) (ex-; prefix written with hyphen but treated as trigger word) |
| dirprwy | SM    | —                                       | §96(c)                                                               |
| uwch    | SM    | —                                       | §96(c)                                                               |
| is-     | SM    | —                                       | §96(c)                                                               |
| cryn    | SM    | —                                       | §96(d)                                                               |
| unig    | SM    | —                                       | §99 (prenominal sense 'only'; postnominal 'lonely' does not trigger) |
| rhyw    | SM    | —                                       | §115                                                                 |
| unrhyw  | SM    | —                                       | §115                                                                 |
| amryw   | SM    | —                                       | §116                                                                 |
| cyfryw  | SM    | —                                       | §116 (y cyfryw °beth)                                                |
| fath    | SM    | —                                       | §116 (y fath °beth)                                                  |
| ffasiwn | SM    | —                                       | §116                                                                 |
| rhai    | none  | —                                       | §115 (contrast rhyw°)                                                |
| sawl    | none  | —                                       | §187                                                                 |
| peth    | none  | —                                       | §193 (directly precedes sg noun, no o°, no mutation)                 |
| dau     | SM    | —                                       | §162(b)                                                              |
| dwyl    | SM    | —                                       | §162(b)                                                              |
| tri     | AM    | —                                       | §162(c) 'erratically in the spoken language' CONTESTED (am-erosion)  |
| tair    | none  | —                                       | §162(c) (always radical)                                             |
| pedwar  | none  | —                                       | §162                                                                 |
| pedair  | none  | —                                       | §162                                                                 |
| chwe    | AM    | —                                       | §162(e) 'erratically in speech' CONTESTED (am-erosion)               |
| chwe    | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod | §176; radical sometimes (chwe blynedd) CONTESTED (nm-blynedd)        |
| pum     | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod | §176                                                                 |
| saith   | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod | §176                                                                 |

| Lemma     | Grade | Target conditions                                                                | Source                                                                 |
|-----------|-------|----------------------------------------------------------------------------------|------------------------------------------------------------------------|
| wyth      | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod                                          | §176; radical sometimes<br>(wyth blynedd)<br>CONTESTED<br>(nm-blynedd) |
| naw       | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod                                          | §176                                                                   |
| deng      | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod                                          | §176                                                                   |
| deuddeng  | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod                                          | §176 (12)                                                              |
| pymtheng  | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod                                          | §176 (15)                                                              |
| deunaw    | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod                                          | §176 (18)                                                              |
| ugain     | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod                                          | §176 (20)                                                              |
| can       | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod                                          | §176 (100: can<br>°mlwydd oed)                                         |
| ail       | SM    | —                                                                                | §170(b) (both genders)                                                 |
| trydedd   | SM    | gender=f                                                                         | §170(b)                                                                |
| pedwaredd | SM    | gender=f                                                                         | §170(b)                                                                |
| trydydd   | none  | —                                                                                | §170(b) (masc: y<br>Trydydd Byd)                                       |
| pedwerydd | none  | —                                                                                | §170(b)                                                                |
| pumed     | SM    | gender=f                                                                         | §170(b) (y °bumed<br>°orsaf; masc radical)                             |
| chweched  | SM    | gender=f                                                                         | §170(b)                                                                |
| seithfed  | SM    | gender=f                                                                         | §170(b)                                                                |
| wythfed   | SM    | gender=f                                                                         | §170(b)                                                                |
| nawfed    | SM    | gender=f                                                                         | §170(b)                                                                |
| degfed    | SM    | gender=f                                                                         | §170(b)                                                                |
| ugeinfed  | SM    | gender=f                                                                         | §170(d) (yr ugeinfed<br>°ganrif)                                       |
| nos       | SM    | lexemes:<br>Llun/Mawrth/Mercher/Iau/ <del>Calan</del> /Mer/Sad-<br>wrn/Sul/Calan | §180 (Nos °Fawrth, Nos<br>Calan)                                       |
| mewn      | none  | —                                                                                | §461 (non-specific ‘in a’;<br>contrast yn.loc)                         |

| Lemma      | Grade | Target conditions                       | Source                                                                                              |
|------------|-------|-----------------------------------------|-----------------------------------------------------------------------------------------------------|
| mo         | SM    | —                                       | §462, §295 (contraction of dim o°, carries o's SM)                                                  |
| oddiar     | SM    | —                                       | §463                                                                                                |
| oddiwrth   | SM    | —                                       | §464                                                                                                |
| rhag       | none  | —                                       | §465 (rhag dod, rhag ofn — radical)                                                                 |
| rhwng      | none  | —                                       | §466 ('one of the few simple spatial prepositions which does not cause SM')                         |
| trw        | SM    | —                                       | §468 (variant of trwy)                                                                              |
| wth        | SM    | —                                       | §470 (spoken variant of wrth)                                                                       |
| er         | none  | —                                       | §506 (er ei holl gyfoeth — no mutation from er itself)                                              |
| ers        | none  | —                                       | §503                                                                                                |
| erbyn      | none  | —                                       | §503                                                                                                |
| nes        | none  | —                                       | §503 (i-construction; no direct mutation)                                                           |
| ar bwys    | none  | —                                       | §475 (compound preps cause no mutation on following noun)                                           |
| ar draws   | none  | —                                       | §475                                                                                                |
| ar gyfer   | none  | —                                       | §475                                                                                                |
| ar ôl      | none  | —                                       | §475                                                                                                |
| er mwyn    | none  | —                                       | §475                                                                                                |
| o flaen    | none  | —                                       | §475                                                                                                |
| o gwmpas   | none  | —                                       | §475                                                                                                |
| wrth ymyl  | none  | —                                       | §475                                                                                                |
| ymhlith    | none  | —                                       | §475-476                                                                                            |
| ymysg      | none  | —                                       | §475-476                                                                                            |
| yn lle     | none  | —                                       | §475                                                                                                |
| hanner can | NM    | lexemes:<br>blwydd/blynedd/di-<br>wrnod | §176 (50)                                                                                           |
| a.conj     | AM    | —                                       | §510; 'often disregarded in the spoken language, particularly for th-'<br>CONTESTED<br>(am-erosion) |
| na.nor     | AM    | —                                       | §513 (nor; 'often disregarded in speech')<br>CONTESTED<br>(am-erosion)                              |



| Lemma    | Grade | Target conditions | Source                                                                                                                         |
|----------|-------|-------------------|--------------------------------------------------------------------------------------------------------------------------------|
| na.rel   | mixed | —                 | §481 (NEG relative; King's exx show SM on b/d/g-initials, consistent with mixed; AM on c/p/t per prescriptive tradition + §10) |
| sy       | SM    | lexemes: dim      | §479 (sy °ddim)                                                                                                                |
| os       | none  | —                 | §502 (os daw e)                                                                                                                |
| pe       | none  | —                 | §502                                                                                                                           |
| hyd.conj | none  | —                 | §502 (hyd gwelwch chi — verb radical; contrast hyd° preposition)                                                               |
| ta       | none  | —                 | §512 (te 'ta coffi)                                                                                                            |
| ond      | none  | —                 | §511                                                                                                                           |
| ni       | mixed | —                 | §10                                                                                                                            |
| na.neg   | mixed | —                 | §10                                                                                                                            |
| oni      | mixed | —                 | §10                                                                                                                            |

## 9. Known limits

- **Adverbial and vocative status are author-supplied.** Geometry provably underdetermines them: sentence-initial adverbial NPs mutate while geometrically identical fronted objects do not. The distinction is adjunct-vs-argument, a valence/lexical fact — every formal account consumes it as input (Dowle 2024 takes it from f-structure), none derives it.
- **Common-noun apposition** shares the genitive configuration and would be misclassified as a possessor; appositive NPs are overwhelmingly personal names (immutable), so the exposure is minimal.
- **Coordination** is a known soft spot for the XPTH generally; Welsh coordinators happen to begin with mutation-immune segments (*a, neu, ond, na*), which masks most of the exposure.
- **Usage variability** (§7) is out of predictive scope by design; a usage model would attach per-trigger application rates to the same rule inventory.
- The mixed-mutation grade of the negative relative particle *na* is encoded from prescriptive tradition; King's own examples (§481) are grade-ambiguous (all on *b/d/g*-initials, where mixed and SM coincide).

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